

Course Code	21CSC574J	Course Name	BIG DATA ANALYTICS USING ARTIFICIAL INTELLIGENCE TECHNOLOGIES	Course Category	C	PROFESSIONAL CORE	L	T	P	C
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Pre-requisite Courses	Nil	Co-requisite Courses	Nil	Progressive Courses	Nil
Course Department	Offering	Computational Intelligence	Data Book / Codes/Standards	Nil	

Course Learning Rationale (CLR)	The purpose of learning this course is to:
CLR-1 :	Understand the basics of Hadoop and related technologies to big data analytics and Examine the ecosystem tools in hadoop
CLR-2 :	Provide with comprehensive Apache Spark & Spark SQL knowledge for Big data processing
CLR-3 :	Provide with comprehensive in Spark MLlib and structured streaming
CLR-4 :	Recognize the applications with Apache airflow and Zookeeper
CLR-5 :	Explore the advantages of using Ontologies for enhanced data integration and knowledge representation in Big Data contexts.

Course Outcomes (CO):	At the end of this course, learners will be able to:	Program Outcomes (PO)		
		1	2	3
CO-1:	Apply MapReduce, HDFS and YARN develop big data applications	2	2	2
CO-2:	Provide with comprehensive Apache Spark & Spark SQL knowledge for Big data processing	2	2	2
CO-3:	Identify the importance of Spark MLlib and structured streaming	2	2	2
CO-4:	Identify the importance of Apache Airflow and Zookeeper tool	2	2	2
CO-5:	Utilize Ontologies to enhance data integration, interoperability, and knowledge representation in Big Data environments.	3	3	3

Module-1 Introduction to HDFS and Hive	12 Hours
Overview of Hadoop- HDFS architecture-Hadoop MapReduce-Processing data with map reduce-YARN architecture-Anatomy of file Read and Write in Hadoop-Hadoop Shell commands - Overview of Hadoop Ecosystem-Apache Hive-Data Types>Create, Alter and Drop Commands-Built in operators and Functions-Views-Interaction with Hadoop framework. Lab 1 : HDFS Shell commands Lab 2 : Job execution on Yarn Lab 3 : HQL executions	
Module -2 Apache Spark and SparkSQL	12 Hours
Introduction to Apache Spark- Architecture of Apache Spark- RDDs (Resilient Distributed Datasets): creation, transformations, and actions-Spark SQL-SQL queries in Spark Lab 4: Creating RDD from external sources Lab 5: Create Dataframe from CSV Lab 6: Dataframe operations	

Module – 3 Spark MLlib and Spark structured Streaming	12 Hours
Spark MLlib – pySpark – Spark Structured Streaming –Basic concepts, Programming Model, APIs using Streaming Dataframes, Creating Streaming Dataframes, Operations using Streaming dataframes Lab 7 : Features Engineering using Spark MLlib Lab 8 : Classification and Regression model building using Spark Lab 9 : Stream processing using Kafka	
Module - 4 Zookeeper and Apache Airflow	12 Hours
Introduction to Apache, Kafka-Cluster Architecture –Kafka messaging –components in messaging – creating and deleting kafkaTopic- Creating multiple brokers- Kafka producers- API for a kafka producers-Producer implementation – Three ways to send kafka message- Kafka consumers – Consumer group- API for kafka Consumer-Consumer Implementation-Integration with Spark, Apache, Airflow- Directed Acyclic Graph – CLI commands Lab 10 : Znode demo Lab 11 : Task scheduling using Airflow Lab 12 : Orchestration demo with Airflow	
Module -5 Ontology for Big Data	12 Hours
Human brain and Ontology, Ontology of information science, Ontology properties, Advantages of Ontologies, Components of Ontologies, The role Ontology plays in BigData, Ontology alignment, Goals of Ontology in big data, Challenges with Ontology in Big Data, RDF – the universal data format, Using OWL, the Web Ontology Language, SPARQL query language, Building intelligent machines with Ontologies, Ontology learning, Ontology learning process. Case studies for Big data using artificial Intelligence Technologies	

Learning Resources	1. Anand Deshpande & Manish Kumar, Artificial Intelligence for Big Data, Packt Publishing, 2018. 2. Rinku Sharma Dixit & Shailee Lohmor Choudhary, Big Data Analytics Using Artificial Intelligence Technologies Transforming Organizations, IK International Pvt. Ltd, 2021 3. Seema Acharya, Big Data and Analytics, Wiley, 2019	4. Raj Kamal and Preeti Saxena, Big Data Analytics: Introduction to Hadoop, Spark, and Machine-Learning, McGraw Hill Education, 2019 5. www.spark.apache.org 6. www.splunk.com 7. Databricks Documentation: https://docs.databricks.com/en/index.html
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Learning Assessment							
Bloom's Level of Thinking		Continuous Learning Assessment (CLA) - By the Course Faculty				By The CoE	
		Formative CLA-I Average of unit test (45%)		Life Long* Learning CLA-II- Practice (15%)		Summative Final Examination (40% weightage)	
		Theory	Practice	Theory	Practice	Theory	Practice
Level 1	Remember	25%	-	-	15%	15%	-
Level 2	Understand	25%	-	-	25%	20%	-
Level 3	Apply	20%	-	-	20%	15%	-
Level 4	Analyze	15%	-	-	20%	25%	-
Level 5	Evaluate	15%	-	-	20%	25%	-
Level 6	Create	-	-	-	-	-	-
	Total	100 %		100 %		100 %	

Course Designers		
Experts from Industry	Experts from Higher Technical Institutions	Internal Experts
1.Mr.KaustuvKunal, <i>Principal Data scientist, Great Learning</i> 2.Mr. S. Smilin Sam, <i>Principal Engineer, AthenahealthPvt Ltd.</i>	1.Dr. R. Gayathri, <i>Professor, Sri Venkateswara College of Engineering, Sriperumbudur.</i>	1.Dr.G.Maragatham , <i>SRMIST</i> 2.Dr. C. SherinShibi, <i>SRMIST</i>